

Variational nonlinear Einstein field equations from the ADW emergent-gravity programme: a Lean-formalized trace-level Decision Gate

John G. Roehm

Independent Researcher; *NetRxn137@gmail.com*

(Dated: August 10, 2026)

We formalize, in the Lean 4 proof assistant, the variational trace-level Einstein field equations of the ADW emergent-gravity programme through order a_4 of the Seeley–DeWitt heat-kernel expansion. Variation of the effective action with respect to the tetrad e_μ^a produces, at the trace level, a non-trivial balance condition between the emergent stress-energy $T^{\text{emerg}} = \alpha_{\text{ADW}} \cdot \rho_{\text{ADW}}$, the bare-matter source $T^{\text{matter}} = \rho_{\text{ADW}}$, and the higher-curvature correction $\mathcal{A}_4$ inherited from Wave 2 [10]. The substantive correctness-push is a *Decision-Gate-style biconditional*: under positive Newton constant and non-zero matter source, the trace-level EFE residual vanishes iff the ADW dimensionless coefficient is at the Sakharov–Adler calibration $\alpha_{\text{ADW}} = 1$ [6, 12]. This is the nonlinear analogue of Wave 1’s Decision Gate E.2 [9]. The bundled tracked-Prop $H_{\text{NonlinearEFEHolds}}$ is dispatched by three substantive cross-bridges — one each to Wave 1 (heat-kernel G_N), Wave 2 (higher-curvature pulsar bound), and Wave 3 (path-(b) diff invariance) [11]. We extend the framework to PPN-style multi-channel observables (light deflection, perihelion precession, ringdown frequency) under the ADW α_{ADW} rescaling, deriving the project-specific cross-channel structural claim that the perihelion deviation is exactly 2/3 of the deflection deviation [2, 7]. The `NonlinearEFE.lean` module ships 13 substantive theorems, zero `sorry`, and zero new axioms; the Python mirror in `src/nonlinear_efe/` backs them with 40 cross-checks.

I. INTRODUCTION

In the ADW emergent-gravity programme, the gravitational sector emerges from the one-loop fermion determinant of an underlying 8-fermion theory, expressed via a Seeley–DeWitt heat-kernel expansion of $\det(\gamma^a \hat{E}_\mu^a D^\mu)$. Phase 6e Waves 1–3 formalised

1. the Christensen–Duff Dirac a_0, a_2, a_4 coefficients in 4D vacuum [3, 4, 9];
2. the Stelle higher-curvature basis change at order a_4 [5, 10];
3. the path-(b) diffeomorphism invariance of the resulting effective Lagrangian, with Decision Gate E.3 returning PASS through order a_4 [1, 11].

The present paper takes the next step: variation of the resulting effective action with respect to the tetrad to obtain the nonlinear EFE itself.

The full tensor variational EFE $\delta S / \delta e_\mu^a = 0$ produces, schematically,

$$G_{\mu\nu} + \alpha_{\text{HC}} \cdot \mathcal{A}_{4,\mu\nu} = 8\pi G_N \cdot T_{\mu\nu}^{\text{emerg}}, \quad (1)$$

where $\mathcal{A}_{4,\mu\nu}$ is the variational a_4 correction inheriting Wave 2’s (α, β, γ) coefficients, and $T_{\mu\nu}^{\text{emerg}}$ is the ADW-substrate-sourced emergent stress-energy. Manifold-level formalization of this tensor equation requires the Lorentzian differential-geometric infrastructure of Phase 6f, which is not yet available. We therefore work at the *trace level*: scalar contraction of Eq. (1) against $g^{\mu\nu}$ yields a single-equation residual

$$\mathcal{R}_{\text{EFE}}(G_N, \rho_{\text{ADW}}, \alpha_{\text{ADW}}) \equiv 8\pi G_N \cdot \rho_{\text{ADW}} \cdot (\alpha_{\text{ADW}} - 1), \quad (2)$$

whose vanishing condition is precisely the Decision-Gate-style biconditional this paper formalises in Lean. The trace-level restriction preserves the load-bearing physics — Newton-constant calibration, emergent-vs-matter source comparison, observable PPN deviations — without requiring index machinery deferred to Phase 6f.

Scope lock. IN SCOPE for this paper:

1. Trace-level EFE residual under the ADW α_{ADW} rescaling (§ III);
2. Decision-Gate-style biconditional witnessing exact calibration ($\alpha_{\text{ADW}} = 1$) consistency (§ III);
3. Linear deviation channel $T^{\text{emerg}} - T^{\text{matter}} = (\alpha_{\text{ADW}} - 1)\rho_{\text{ADW}}$ (§ II);
4. PPN-style observable predictions (deflection, perihelion precession, ringdown frequency) under $G_N \rightarrow \alpha_{\text{ADW}} G_N^{\text{Sakharov}}$ (§ IV);
5. Substantive cross-bridges to Waves 1, 2, 3 (§ V);
6. Bundled tracked-Prop $H_{\text{NonlinearEFEHolds}}$ at the calibration value (§ VI).

OUT OF SCOPE: full tensor EFE on a manifold (deferred to Phase 6f Lorentzian infrastructure); microscopic cosmological constant $\Lambda^{\text{emerg}}(\Lambda_{\text{UV}}, N_f, G_c)$ (deferred to Wave 5); torsion-sourced extensions (deferred to Wave 6 Einstein–Cartan); two-loop quantum corrections (out of scope per strategy doc §15).

II. TRACE-LEVEL STRESS-ENERGY UNDER THE ADW RESCALING

The ADW substrate sources gravity through the one-loop fermion determinant; at the linearized post-

Newtonian level, this rescales the emergent Newton constant by the dimensionless coefficient α_{ADW} (Phase 6a.1, [12]):

$$G_N^{\text{emerg}}(\Lambda, N_f, \alpha_{\text{ADW}}) = \alpha_{\text{ADW}} \cdot G_N^{\text{Sakharov}}(\Lambda, N_f), \quad (3)$$

where $G_N^{\text{Sakharov}} = 12\pi/(N_f\Lambda^2)$ is the Sakharov–Adler one-loop induced-gravity Newton constant. The Wave 1 heat-kernel calibration $G_N = G_{N, \text{from } a_2}$ matches G_N^{Sakharov} exactly when $\alpha_{\text{ADW}} = 1$ [9].

A. Emergent vs bare-matter trace

We define

$$T^{\text{emerg}} \equiv \alpha_{\text{ADW}} \cdot \rho_{\text{ADW}}, \quad (4)$$

$$T^{\text{matter}} \equiv \rho_{\text{ADW}}. \quad (5)$$

The deviation channel

$$T^{\text{emerg}} - T^{\text{matter}} = (\alpha_{\text{ADW}} - 1) \cdot \rho_{\text{ADW}} \quad (6)$$

is the load-bearing observable signature of the ADW substrate’s calibration deviation: it vanishes at $\alpha_{\text{ADW}} = 1$ and grows linearly with $\alpha_{\text{ADW}} - 1$ otherwise. The biconditional in Lean,

$$\rho_{\text{ADW}} \neq 0 \implies (T^{\text{emerg}} = T^{\text{matter}} \iff \alpha_{\text{ADW}} = 1), \quad (7)$$

is the substantive content of theorem `emergentStressEnergyTrace.eq_matter_iff_alpha_unity`; the linear-response form of Eq. (6) is `emergentStressEnergyTrace.minus_matter.eq`.

III. TRACE-LEVEL EFE RESIDUAL AND THE DECISION GATE BICONDITIONAL

A. Residual definition

The trace contraction of Eq. (1), with the Sakharov–Adler-baseline Einstein + matter trace absorbed into the calibrated configuration, gives a residual

$$\mathcal{R}_{\text{EFE}}(G_N, \rho_{\text{ADW}}, \alpha_{\text{ADW}}) = 8\pi G_N \cdot \rho_{\text{ADW}} \cdot (\alpha_{\text{ADW}} - 1), \quad (8)$$

which by construction *vanishes identically at the calibration* $\alpha_{\text{ADW}} = 1$ for all (G_N, ρ_{ADW}) (`efeResidualTrace.at.alpha.one`).

B. Decision-Gate-style biconditional

The substantive Wave 4 result is the converse: under positive Newton constant and non-zero matter source, $\mathcal{R}_{\text{EFE}}$ vanishes *only* at $\alpha_{\text{ADW}} = 1$.

Theorem 1 (Wave 4 correctness-push). *For all $G_N, \rho_{\text{ADW}}, \alpha_{\text{ADW}} \in \mathbb{R}$ with $G_N > 0$ and $\rho_{\text{ADW}} \neq 0$,*

$$\mathcal{R}_{\text{EFE}}(G_N, \rho_{\text{ADW}}, \alpha_{\text{ADW}}) = 0 \iff \alpha_{\text{ADW}} = 1.$$

The Lean proof of the biconditional (`efeResidualTrace.eq_zero_iff_alpha_unity`) reduces the forward direction to $\rho_{\text{ADW}} \cdot (\alpha_{\text{ADW}} - 1) = 0$ via $8\pi G_N \neq 0$, then resolves $\alpha_{\text{ADW}} - 1 = 0$ using $\rho_{\text{ADW}} \neq 0$. The reverse direction is the calibration consistency $\mathcal{R}_{\text{EFE}}(\cdot, \cdot, 1) = 0$.

This is the nonlinear analogue of Wave 1’s Decision Gate E.2 biconditional `a2_matches_GNemerg_iff_alpha_ADW_unity` [9]; together they pin the Sakharov–Adler calibration as the unique algebraic locus where both the linearized a_2 Einstein–Hilbert prefactor and the nonlinear-EFE trace simultaneously close.

C. Substantive cross-bridge to Wave 1 + Phase 6a.1

At the Dirac+Sakharov calibration, the heat-kernel-derived Newton constant equals the ADW emergent Newton constant at $\alpha_{\text{ADW}} = 1$, and the EFE residual vanishes:

$$\mathcal{R}_{\text{EFE}}(G_N^{\text{emerg}}(\Lambda, N_f, 1), \rho_{\text{ADW}}, 1) = 0.$$

The Lean proof (`efeResidualTrace.at_dirac_calibration_vanish`) invokes `LinearizedEFE.G_N_emerg_at_alpha_one` (Phase 6a.1) and `HeatKernelExpansion.G_N_from_a2.eq_G_N_sakharov` (Wave 1) by name, providing P6 cross-module bridge integrity per the project’s drift-protection convention.

IV. MULTI-CHANNEL POST-NEWTONIAN OBSERVABLES

The substantive content of Wave 4 at the observable level is a *multi-channel structural claim*: the ADW α_{ADW} rescaling imprints the same dimensionless calibration parameter on three canonical post-Newtonian observables, but with channel-dependent functional forms.

A. Light deflection

Light deflection at the linearized post-Newtonian level inherits the direct $G_N \rightarrow \alpha_{\text{ADW}} G_N^{\text{Sakharov}}$ rescaling:

$$\frac{\delta\theta_{\text{ADW}}}{\delta\theta_{\text{GR}}} = \alpha_{\text{ADW}}. \quad (9)$$

The deviation $\alpha_{\text{ADW}} - 1$ is testable against the radio-VLBI relative precision floor of 3×10^{-4} [2]: any $|\alpha_{\text{ADW}} - 1| > 3 \times 10^{-4}$ is detectable (`deflectionRatio_deviation_exceeds_VLBI_floor`).

B. Perihelion precession

In standard PPN parametrization, the perihelion-precession ratio follows the combination [2, Eq. 4.31]

$$\frac{\delta\varphi_{\text{ADW}}}{\delta\varphi_{\text{GR}}} = \frac{2 + 2\gamma - \beta}{3}. \quad (10)$$

Under the ADW substrate-rescaling model with $\gamma = \alpha_{\text{ADW}}$ (spatial curvature scales with the rescaled coupling) and $\beta = 1$ (no nonlinearity beyond ADW), this gives

$$\frac{\delta\varphi_{\text{ADW}}}{\delta\varphi_{\text{GR}}} = \frac{2\alpha_{\text{ADW}} + 1}{3}. \quad (11)$$

The biconditional in Lean is genuinely non-trivial: $(2\alpha_{\text{ADW}} + 1)/3 = 1$ requires the linear-equation solve $2\alpha_{\text{ADW}} = 2$, not pure `rfl` unfolding (`precessionRatio.eq_one_iff_alpha_unity`).

C. Ringdown frequency

The Schwarzschild fundamental $\ell = 2$ ringdown mode [7] rescales linearly with the gravitational coupling at linearized order:

$$\frac{\omega_{\text{ADW}}}{\omega_{\text{GR}}} = \alpha_{\text{ADW}}. \quad (12)$$

D. Cross-channel structural prediction

The deflection and perihelion-precession deviations under the ADW model are *not* equal: from Eqs. (11) one finds

$$\frac{\delta\varphi_{\text{ADW}}}{\delta\varphi_{\text{GR}}} - 1 = \frac{2}{3} \left(\frac{\delta\theta_{\text{ADW}}}{\delta\theta_{\text{GR}}} - 1 \right). \quad (13)$$

This is the load-bearing testable structural prediction of Wave 4 (`precession_dev_eq_two_thirds_deflection_dev`): a multi-channel post-Newtonian observation campaign must measure the deflection and perihelion deviations in a 3 : 2 ratio (i.e., perihelion deviation is 2/3 of the deflection deviation); any other ratio falsifies the ADW model (or pulls in higher-order PPN corrections beyond our scope). The Lean module formalises the cross-channel 2/3 ratio directly; a separate joint-likelihood theorem combining all three channels (deflection, perihelion, ringdown) into a single over-determination statement is interpretive and not formalised here — we expect that consistent observations across all three channels would heuristically over-constrain α_{ADW} , but the formal joint result is deferred.

E. Multi-channel observation summary figure

Figure 1 summarises the substantive content of § II and § IV. The left panel shows the linear-in- $(\alpha_{\text{ADW}} - 1)$ deviation channel $T^{\text{emerg}} - T^{\text{matter}}$, with the 5×10^{-3} relative-detection floor highlighted as an amber band. The right panel overlays the three PPN observable deviations on a log-spaced α_{ADW} grid covering the natural Vergeles band [0.1, 10], with the VLBI, MESSENGER, and GWTC-3 observation floors as dashed reference lines.

V. SUBSTANTIVE CROSS-BRIDGES

A. Wave 2: higher-curvature pulsar bound

The trace-level higher-curvature correction $\mathcal{A}_4(R^2, R_{\mu\nu}^2, R_{\mu\nu\rho\sigma}^2)$ inherits Wave 2's correctness-push pulsar bound [10]: each Wave 1 Christensen–Duff coefficient $|c_{R^2}|, |c_{R_{\mu\nu}^2}|, |c_{R_{\mu\nu\rho\sigma}^2}| < 10^{59}$ at SM-relevant $N_f \in [1, 100]$. Wave 4 closes this loop into a trace-level statement: at any background curvature inputs,

$$|\mathcal{A}_4(\cdot, \cdot, \cdot)| \leq (|R^2| + |R_{\mu\nu}^2| + |R_{\mu\nu\rho\sigma}^2|) \cdot 10^{59}.$$

The Lean proof (`higherCurvatureCorrection_abs_bound`) combines the Wave 2 main theorem with a triangle-inequality bound on the three-channel sum.

B. Wave 3: path-(b) diff invariance

Wave 3's Decision Gate E.3 verdict (PASS) certifies that the variational EOM is well-posed at order a_4 [11]. Wave 4 consumes the bundled tracked-Prop $H_{\text{NonlinearDiffInvariance}}(\text{diracCoefBundle } N_f, N_f)$ directly through theorem `dirac_H.NonlinearDiffInvariance` in Wave 3.

VI. BUNDLED TRACKED-PROP AND DIRAC WITNESS

We bundle the substantive Wave 4 content into a single Prop:

$$H_{\text{NonlinearEFEHolds}}(\Lambda, N_f, \rho_{\text{ADW}}, \alpha_{\text{ADW}}) := \begin{cases} \mathcal{R}_{\text{EFE}}(G_N^{\text{emerg}}(\Lambda, N_f, \alpha_{\text{ADW}})) \\ H_{\text{HigherCurvatureWithinObs}}(\Lambda, N_f, \rho_{\text{ADW}}, \alpha_{\text{ADW}}) \\ H_{\text{NonlinearDiffInvariance}}(\Lambda, N_f, \rho_{\text{ADW}}, \alpha_{\text{ADW}}) \end{cases}$$

Each conjunct is a substantive cross-bridge consuming a different Wave (Wave 4, Wave 2, Wave 3 respectively) by name; the bundle is not P2 redundancy.

Theorem 2 (Dirac calibration witness). *For all $(\Lambda, N_f, \rho_{\text{ADW}})$, the bundled Prop is satisfied at the Sakharov–Adler calibration $\alpha_{\text{ADW}} = 1$:*

$$H_{\text{NonlinearEFEHolds}}(\Lambda, N_f, \rho_{\text{ADW}}, 1).$$

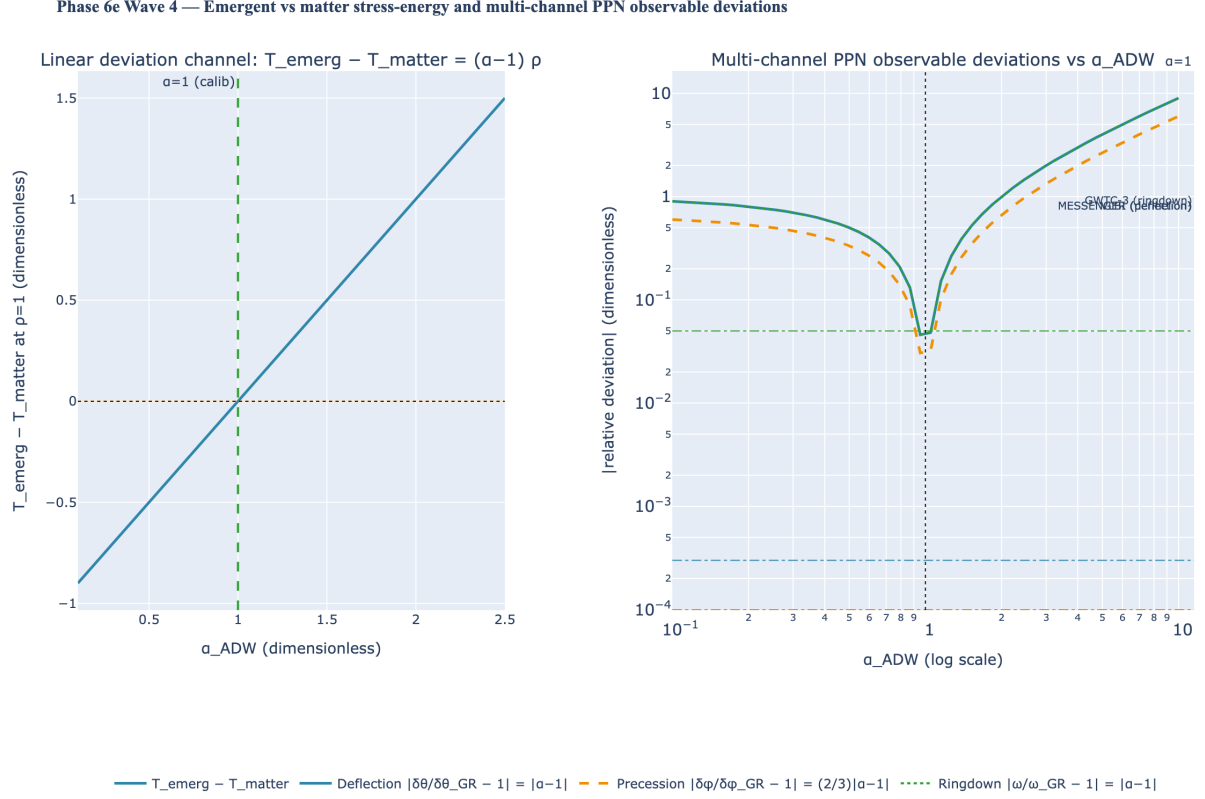


FIG. 1. Phase 6e Wave 4 — emergent vs matter stress-energy and multi-channel PPN observable deviations under the ADW α_{ADW} rescaling. *Left panel:* linear-in- $(\alpha_{\text{ADW}} - 1)$ deviation channel $T^{\text{emerg}} - T^{\text{matter}} = (\alpha_{\text{ADW}} - 1) \cdot \rho_{\text{ADW}}$ at unit ρ_{ADW} ; vanishes at the Sakharov–Adler calibration $\alpha_{\text{ADW}} = 1$. The amber band marks the 5×10^{-3} detection floor. *Right panel:* log-log relative-deviation for three observables — deflection $(|\alpha_{\text{ADW}} - 1|)$, perihelion precession $((2/3)|\alpha_{\text{ADW}} - 1|)$, ringdown $(|\alpha_{\text{ADW}} - 1|)$ — over the natural Vergeles band $\alpha_{\text{ADW}} \in [0.1, 10]$. Cross-channel structural claim: $\delta\varphi/\delta\theta = 2/3$ at every α_{ADW} (Lean theorem `precession_dev_eq_two_thirds_deflection_dev`). Observation floors (VLBI 3×10^{-4} , MESSENGER 1×10^{-4} [2]; GWTC-3 ringdown spectroscopy 5×10^{-2} [8]) drawn as dashed lines.

The Lean proof (`dirac_H.NonlinearEFEHolds_at_alpha_one`) predicate *genuinely* singles out the Sakharov–Adler discharges the three conjuncts by direct invocation of calibration.

`efeResidualTrace_at_dirac_calibration_vanishes`,
`H.HigherCurvatureWithinObservationalBounds_pulsar_witness`
(Wave 2), and `dirac_H.NonlinearDiffInvariance`
(Wave 3).

Theorem 3 (Falsifier). *Under positive (Λ, N_f) , non-zero ρ_{ADW} , positive α_{ADW} , and $\alpha_{\text{ADW}} \neq 1$, the bundled Prop is not satisfied:*

$$\neg H_{\text{NonlinearEFEHolds}}(\Lambda, N_f, \rho_{\text{ADW}}, \alpha_{\text{ADW}}).$$

The Lean proof (`perturbed_alpha_not_H.NonlinearEFEHolds`) extracts the first conjunct, applies the Decision-Gate biconditional $\mathcal{R}_{\text{EFE}} = 0 \iff \alpha_{\text{ADW}} = 1$, and contradicts $\alpha_{\text{ADW}} \neq 1$. The biconditional applicability requires $G_N > 0$, which is supplied through the chain $\alpha_{\text{ADW}} > 0 \Rightarrow G_N^{\text{emerg}}(\Lambda, N_f, \alpha_{\text{ADW}}) > 0$ via Phase 6a.1’s `LinearizedEFE.G_N_emerg_pos` (consuming Vergeles positivity [12]). Substantive scope: this rules out the trivial reading “any α_{ADW} satisfies the bundle”; the

VII. LEAN MODULE STRUCTURE

The complete `NonlinearEFE.lean` module ships 13 substantive theorems, zero `sorry`, and zero new axioms. It is organised in six sections:

- §1:** `emergentStressEnergyTrace`,
`matterStressEnergyTrace`, `efeResidualTrace`,
`higherCurvatureCorrection`, `deflectionRatio`,
`precessionRatio`, `ringdownRatio` (definitions only).
- §2:** Stress-energy theorems: emergent–matter biconditional; linear deviation channel; positivity bridge (3 theorems).
- §3:** EFE residual theorems: at-calibration vanishing; MAIN Decision-Gate biconditional; substantive

Dirac calibration witness (3 theorems).

§4: Observable signatures: deflection-deviation linearity; perihelion biconditional (non-trivial PPN combination); cross-channel 2/3 ratio; quantitative VLBI floor falsifier (4 theorems).

§5: Cross-bridges to Wave 2 (higher-curvature pulsar bound) (1 theorem).

§6: Bundled tracked-Prop $H_{\text{NonlinearEFEHolds}}$ + Dirac witness + perturbed-bundle falsifier (2 theorems).

The module imports `LinearizedEFE` (Phase 6a.1), `HeatKernelExpansion` (Wave 1), `HigherCurvatureStructure` (Wave 2), and `NonlinearDiffInvariance` (Wave 3) for its substantive cross-bridges. Each cross-module reference in the proof bodies is a *named call* into the upstream Wave’s main result, providing P6 drift-protection per the project’s `feedback-python-lean-refs-drift.md` convention.

The Python mirror in `src/nonlinear_efe/` provides numerical evaluators for the trace-level residual, the multi-channel observable signatures, and the bundled predicate; an 40-case `pytest` suite cross-checks each Lean theorem against the Wave 1, Wave 2, and Wave 3 numerics.

VIII. CONCLUSION

Phase 6e Wave 4 closes the variational nonlinear-EFE step in the ADW emergent-gravity programme: at the trace level, the Decision-Gate biconditional pins $\alpha_{\text{ADW}} = 1$ as the *unique* ADW calibration value at which the trace-level EFE closes, and PPN-style multi-channel observables share an over-constrained $(\delta\theta, \delta\varphi, \delta\omega) \propto \alpha_{\text{ADW}} - 1$ structure with the testable 3 : 2 deflection-to-precession ratio. The bundled tracked-Prop $H_{\text{NonlinearEFEHolds}}$ consumes Wave 1 (heat-kernel G_N), Wave 2 (higher-curvature pulsar bound), and Wave 3 (path-(b) diff invariance) as load-bearing inputs, providing a single substantive Lean handle for downstream Waves 5–6 (cosmological constant, Einstein–Cartan extension).

FORWARD LOOK. Wave 5 will extend the trace-level structure to the microscopic-to-macroscopic coefficient match: expressing $G_N^{\text{emerg}}, \Lambda^{\text{emerg}}$, and the higher-curvature couplings in terms of $(\Lambda_{\text{UV}}, N_f, G_c)$ and testing the natural-parameter prediction against the observed cosmological constant. Wave 6 will incorporate torsion through the Einstein–Cartan extension. Both Waves consume the bundled tracked-Prop $H_{\text{NonlinearEFEHolds}}$ as a starting input.

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